Whole-brain CT-MHSA: why the loop gain G is unidentifiable

Intermediate report — 2026-09-17

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Bottom line

1. Two of the three problems were BUGS, not science: the trainer had no functional coupling at all, and the metric was scoring startup transients.
2. The loop gain G is structurally unidentifiable under BPTT: it acts only as a scalar gain on the drive, and this readout+loss pipeline discards amplitude.
3. Proposal: fit G and sigma on the FC metric by an outer loop; keep BPTT for the shape controls (W_q, W_y, kv_scale, tau). Running now.

All numbers on the following slides are reproducible from committed code; the certified metric rule is defined on slide 5.

The model and the fit

Per integration step (dt_train = 100 ms, window = 64 TR, BPTT with remat):

```
aff_t  = G * (C @ y_{t-1})       C = structural connectome (spectral-radius 1)
x_t    = aff_t + noise_t         OU/AR(1) noise, stationary std sigma
M     += u (v k^T - M)           u = dt/tau;  memory (per region, per head)
y      = W_Y M A phi(q);  y_coupled = y / (ema_rms(y) + 1e-8)
e_t    = ||Delta M_t||_F  averaged over heads     ('surprise', >= 0)
```

Training loss per-region z-scored MSE between TR-averaged e and empirical BOLD
Evaluation Pearson corr between simulated and empirical FC matrices (fc_match)
after a Windkessel balloon, scored on a settled window

Tunable parameters: W_q, W_y, kv_scale, log_tau, log_sigma, log_noise_tc, and G (the loop gain).

Bug 1 — the model had no coupling

The reference simulator couples regions through the EMA-normalised output:

```
wholebrain_fc.py:   y_out = y / (y_ema + 1e-8);   history.at[t % L].set(y_out)
```

The trainer computed the identical quantity and then discarded it:

```
wb_train.py:        y_out = y / (y_ema + 1e-8)
                    return (state.M, noise, y_ema, y), e      # couples through raw y
```

With kv_scale collapsed to 0.0032, raw y has rms ~5.7e-09 -> the afferent drive is ~3.4e-09 against noise 0.145 (SNR ~2e-08). The 'whole-brain network' was 100 independent noise-driven blocks, and fc1 = 0.209 came from the SHARED random projection matrices, not from coupling.

Fix (--coupling ema + persistent carry): both attribution factors rose ~300x. kv_scale then stopped collapsing: 0.0032 -> 0.40-0.65.

Bug 2 — the metric was measuring startup transients

Legacy fc_match integrated 600 s from the default balloon initial condition and scored the WHOLE trace. Transient contamination, first-10-TR std / resting std: 103x - 541x in 6 of 7 arms; only the baseline (1.5x) and one arm (5.4x) were clean.

Settled-window cap bound in 5/12 subject-runs at 1200 s and 8/12 at 2400 s -> those windows were truncated, not certified settled.

arm (legacy: 600 s whole trac	fc1	fc2
wb_train baseline	+0.209	+0.118
r3_fc2	+0.224	+0.180
r3_fcy300	+0.172	+0.106

Certified rule (used for all comparisons from here):
2400 s sim; 10-TR rolling std; ref = median over the final 50% of the trace;
W = first TR starting 50 consecutive TRs within 10% of ref; NO cap; FC on bold[W:].

The ranking inverted under the certified metric

arm (certified: 2400 s, strict settle, no	fc1	fc2_holdout
ema_yout — loss=current, ema coupling, p	+0.081	+0.179
fcy_noreg — FC-shaped loss on y-traces	+0.114	+0.021
current_logg — log-G per-region MSE	-0.082	+0.210
r3_fc2 — combined y+e FC loss	+0.035	+0.066
r3_fcy300 — FC loss, 300 iters	-0.010	+0.002
wb_train — original baseline (raw coupling)	-0.049	-0.007

r3_fcy300 was our 'best quality' arm on legacy numbers (fc1 0.172) and collapses to -0.002 once settled; r3_fc2's lead was a cap artefact (W = 54 TR at 1200 s).
The robust winner is the SIMPLEST recipe: the original per-region MSE loss with ema coupling + persistent carry. The raw-coupling baseline is last.

Instrument: exact attribution of dL/dG

G enters at exactly one place per step (aff_t = G * C @ y_{t-1}), so
dL/dG = sum_t < dL/daff_t , daff_t/dG >
with aff injected as an explicit scan input (reverse demand exact per step) and daff/dG obtained by jax.jvp (forward supply). The identity reproduces dL/dG to 9 significant digits in every configuration -> the decomposition is exact.

checkpoint	dL/dG	||dL/daff||	||daff/dG||	null_frac	dL/dG no z-score
ema_yout (burn-in 200)	-5.0e-06	0.044	0.076	0.966	-1.8e-07
ema_yout (burn-in 0)	-2.1e-06	0.043	0.054	0.964	-1.1e-07
baseline wb_train	-3.7e-12	1.4e-04	9.2e-06	0.972	-4.6e-15
drive-norm fix (falsified)	+1.1e-13	1.4e-04	2.3e-05	0.968	+4.6e-15

Both factors are healthy at the drive (0.044 and 0.076) — so it is neither that the loss ignores the drive, nor that G fails to move it.

Where the identifiability is lost

stage	dlog‖stage‖ / dlogG
aff (the coupling drive)	1.0000
x = aff + noise (block input)	6.2e-04
e (surprise readout)	3.0e-04
e_vol (TR-averaged)	2.6e-04
y (block output)	1.4e-03
M (memory)	1.2e-03

G controls aff perfectly, but x = aff + noise only 6.2e-04 — the drive is 4-5 orders of magnitude below the noise floor, so the dilution happens at the sum, and everything downstream inherits that factor.

Of what survives, 96.6% lies in the null space of the per-region z-score (span{1, e_r} per region), and the residual cancels in time: the per-chunk contributions alternate (+2.8e-6, +1.8e-6, -2.4e-6, -1.8e-6, +2.7e-7, +5.3e-6, -1.0e-5, -5.8e-7) and sum to ~5e-6. The recurrent path transmits ~nothing (direct -5.1e-6 vs recursive +4.5e-8).

The mechanism: G is a pure gain, and amplitude is discarded

aff = G * (C @ y_prev): G scales the drive's MAGNITUDE but never its DIRECTION.
So G can only be seen through amplitude — and this pipeline throws amplitude away
three times over:

(a) noise dilution drive rms 7e-06 vs noise 0.145 (SNR ~5e-05)
(b) scale-free loss the z-scored MSE is invariant to per-region scaling
(c) any normalisation dividing the drive by its own rms removes G entirely

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Prediction and test: normalising the drive (the fix designed to raise SNR) should
destroy G — and it did. log-sensitivity of x fell from 6.2e-04 to 1.4e-07, and
dL/dG went from -5e-06 to +1e-13. Removing the z-score ALSO made it smaller
(-1.8e-07), so the normaliser is not the culprit: the amplitude channel itself is.

Dead ends (recorded so they are not retried)

balloon initial conditions	nuisance: they touch only the first ~30 s (frozen-tail max |dBOLD| ~ 8e-06); per-region ICs inflate fc2 from 0.118 to 0.623 with no model change
FC-shaped losses (fcy/fce/fc2)	no robust gain under the certified metric
pulse excitation	no effect on the certified score
log-G reparameterisation	G moves, but that is not identifiability
drive normalisation	falsified its own hypothesis (slide 8)
normalisation asymmetry	real, but not causal (removing the z-score hurts)

Consistent with the reference pipeline: wholebrain_fc.py never learns G — it SWEEPS G and selects by fc_match, a stationary, amplitude-sensitive statistic.

Two-stage test: stage 1 (G, sigma on FC, no gradients)

Shape parameters frozen from ema_yout; G in {0.5 ... 4.0} x sigma in {0.05 ... 0.30}.
Search scored on a settled window (600 s, 20% rule); top 5 re-scored with the
certified rule (2400 s, strict, no cap).

(G, sigma)	lenient fc1 (search)	certified fc1	certified fc2
2.0 / 0.05	0.298	-0.046	+0.101
4.0 / 0.15	0.254	-0.019	+0.084
2.0 / 0.10	0.227	+0.012	+0.026
2.5 / 0.10	0.217	-0.008	+0.047
3.0 / 0.15	0.206	+0.021	-0.041

Sensitivity (certified fc1, subj1) at sigma = 0.15 across G:
    0.5 +0.074 | 1.0 +0.085 | 1.5 +0.098 | 2.0 +0.074 | 2.5 +0.023 | 3.0 +0.021 | 4.0 -0.019
    sd 0.040, span 0.117 -> fc_match is NOT flat in G (3-4x the seed scatter)
At G = 3.0 across sigma: sd 0.042, span 0.129 -> sigma is observable too.

BUT the holdout subject peaks at G = 4, not at subj1's G = 1.5: the metric MOVES G,
it does not PIN it. And the lenient search ranks the certified winner LAST.

Stage 1 took ~1 h on one GPU; the sweep is cheap because it needs no gradients.

Two-stage test: stage 2 and verdict (NO)

arm	final loss	frozen G / sigma	cert fc1	cert fc2	mean
BPTT, seed 1234	1.998	1.5 / 0.15	+0.017	+0.214	+0.116
BPTT, seed 7	2.003	1.5 / 0.15	-0.015	+0.043	+0.014
ema_yout (re-scored)	-	2.23 / 0.19	+0.081	+0.179	+0.130
stage-1 only (no BPTT)	-	1.5 / 0.15	+0.098	+0.026	+0.062

Verdict: the two-stage split does NOT beat ema_yout (best arm mean 0.116 vs 0.130), and both BPTT arms collapse on subj1 (fc1 +0.017 / -0.015). Freezing G is sound, but choosing it on FC is not: the optimum flips between subjects, so the choice is a coin flip on the holdout. Stage-1-only (no BPTT) is worse still (mean 0.062).

What it DOES establish: fc_match is genuinely sensitive to G and to sigma, so the loop gain is not a pure nuisance parameter -- it is under-determined by a single subject's FC, not invisible. That is the thread worth pulling next.

The simulated FC is a global mode — and it is not the noise

Diagnostic over the cohort (job 4828980): the simulated FC stays almost rank-1 under EVERY noise intervention tried, so the 'the noise is i.i.d. per region' explanation is dead.

condition (all at G = 4.	eig_sim top-1	eig_emp top-1	within-max	block_corr	fc
sigma = 0.05 (anchor)	97.9	18.5	0.997	0.352	0.102
sigma = 0.30	96.3	19.2	0.986	0.175	0.100
sigma = 1.00	95.5	19.2	0.985	0.012	0.019
baseline-std variant	99.3	19.2	0.998	0.432	0.208
noise variant 2	94.8	19.2	0.997	0.349	0.125
region-scaled empirical n	99.1	19.2	0.999	0.455	0.192

In every condition the readout e is 92-99% correlated ACROSS REGIONS, and cos(x, aff) ~ 0.85: the readout carries almost no region-specific information. The readout common mode is the strongest remaining candidate, but it is entangled with the coupling drive.

Untested and decisive: does LOWERING G grade the spectrum? Filed as the next card.
Also note the trap this exposed: the baseline-std and empirical-noise conditions RAISE the scalar (fc 0.208, block_corr 0.46-0.58) while leaving the rank-1 structure untouched — exactly the failure mode the low-rank diagnostics exist to catch.

Status, caveats, next steps

Established: the machinery is correct and the metric is fair; the certified winner is ema_yout (fc1 +0.081 / fc2 +0.179); G's unidentifiability has a measured mechanism.

Caveats

- the fits are still weak: best certified number is ~0.18 on one subject;
- some certified windows still hit the cap (noted per-arm) and are not certified settled;
- the winning recipe is n=3 seeds with a wide spread (legacy fc1 0.078 - 0.301);
- attribution used one window per config (three configs agreed).

Next

1. look at the REGIONAL structure, not the scalar: fc_match is one number over 4950 edges, and a 0.2 upper-triangle correlation can hide good structure;
2. fit G/sigma with BOTH subjects in the objective (the flip says one subject under-determines G) -- or report G as a range, not a point;
3. more seeds on the ema_yout recipe to bound its spread;
4. retire the dead knobs; register the certified metric as the standard.